[Github](#) | [Figma](#) | [Group Documentations](#) | [Cohort 7 Deliverables](#)

Vision

SolarEase is a multi-community platform that empowers local environmentally disadvantaged communities to make informed and collective decisions about community solar trade with corporations (like Microsoft), guided by an embedded AI agent/agentic workflow.

The platform translates complex economic, policy, and energy data into understandable insights, enabling local communities to participate in renewable energy markets and negotiate on more equal footing with large corporate buyers.

Problem Statement

Local disadvantaged communities lack accessible, data-driven tools to understand solar investment returns, collaborate with neighboring areas, and negotiate fair renewable energy agreements with corporate buyers.

In the context of SolarEase, “community” refers to both:

1. Geographic communities: groups of residents or organizations located within a shared physical boundary, such as rural towns, counties, or Electric Membership Corporation (EMC) service areas.
2. Relational communities: networks of individuals or institutions connected by shared identity or affiliation, such as faith-based groups, farmer cooperatives, alumni associations, or non-profit partnerships.

We start with geographic communities because all core feasibility, siting, and policy components depend on physical boundaries. This establishes a solid foundation we can later extend to relational communities once the core platform is validated.

Objectives

Their main goal is to make better, collective energy decisions by using data they can actually understand.

Specifically:
Inform - Access transparent info on cost, ROI, and policy incentives.
Collaborate - Compare with neighboring communities and explore joint projects.
Negotiate - Use aggregated data and AI insights to negotiate directly with corporations (e.g., Microsoft).

Objective	What It Means	How We Achieve It (Features / Design)
Inform	Help communities understand cost, ROI, and	- **Dashboard modules** showing visualized ROI, payback, benefits, and project feasibility scores.

	incentives in plain language.	- **AI Agent** “explain” mode that converts technical terms (e.g., PPA rate, IRR, REC) into human language and provides contextual coaching. - **ROI simulator**: “What if we install 200kW instead of 100kW?” - (Optional) **Policy & Market Feed**: powered by AI to summarize relevant changes (e.g., tax credit updates, RED prices, or material tariffs)
Collaborate	Enable multiple communities to compare data and explore joint projects.	- **Multi-community Collaboration Tool:** map view showing energy potential, costs, and distance between communities. - **AI matchmaking agent**: suggests potential neighboring partners based on complementary energy profiles. - (Optional) **Group simulation feature**: simulate combined ROI if two or more communities partner.
Negotiate	Empower communities to negotiate fairly with corporate buyers.	- **AI negotiation assistant**: Supports individual households within a geographic community by helping them understand their options and express preferences during internal community decision-making. - **Solar Energy Trade Optimization Tools**: Provides data-driven visual comparisons of different corporate offers, grounded in the community’s location, parcel characteristics, potential solar output, and incentive landscape. - **Contract summary generator**: Translates proposed corporate-community agreements into concise, plain-language summaries. - (Optional) **Capital Project Dashboard & Impact Visualization**: track project progress, REC delivery, and social impact (MW installed, etc)

Target User

Primary User: Community Energy Leaders (representative composite of Quincy, East Wenatchee, and Malaga residents)

Demographics:

- Location: Central Washington: rural to semi-rural areas near Microsoft's data centers (Grant and Chelan Counties)
- Population Size: 2,000-14,000[1,2,3] residents per community
- Primary Occupations: Agriculture (orchards, wineries, farming), local services, small business owners, community colleges
- Income Level: Lower-middle to middle income; many qualify as environmentally disadvantaged under DOE/Justice40 criteria
- Infrastructure Context:
    - Good sunlight potential and open land for solar installations
    - Aging grid infrastructure, limited access to capital for clean energy investment
    - Existing large corporate presence (Microsoft's data centers) but low local benefit

Secondary User:

- Electric Membership Corporations (EMCs): Operate as intermediaries purchasing power from larger utilities and distributing to rural members. They can use SolarEase for parcel-level feasibility, rate modeling, and collective negotiation.
- Relational Communities: Non-geographically bound networks such as faith-based groups, farmer cooperatives, alumni associations, and NGOs seeking to co-invest or co-own community solar projects.
- Corporate and NGO Stakeholders: Buyers, sponsors, and advocacy organizations tracking regional clean-energy impact, transparency, and Justice40 alignment.

Needs & Pain Points:

1. Access to understandable energy data
   "We don't have in-house expertise to understand energy economics or policy incentives."
   → plain-language tools to visualize costs, benefits, and subsidies

2. Trust & transparency in corporate deals
   "We don't know if partnering with Microsoft or utilities will actually benefit us."
   → Clear after-incentive ROI breakdowns, contract transparency, and community-first negotiation frameworks

3. Financial empowerment
   Jeremy Smithson:
   - Financial barriers are real; incentives drive adoption
   - Utilities & equity programs are growing but poorly communicated

   → Joint or community-based funding options, grants, or shared ownership models

4. Collaboration opportunities
   "We're stronger together. Maybe we can share a solar field with our neighboring towns."
   → Tools to coordinate across nearby communities and compare potential joint benefits

Wants & Aspirations:

- To gain agency in the clean energy transition, not just be “sites” for corporate installations.
- To understand and decide collectively, with data visualized in a way non-experts can use.
- To generate steady local revenue or cost savings through fair solar participation.

User Quote:
“We’ve watched those giant solar farms pop up across the valley, but none of us knew how to get involved. If this tool can show us what’s possible - and make sure we’re not left out - we’d finally have a seat at the table.”

MVP Feature List

1. ROI Analysis Dashboard

A simple, interactive dashboard showing:
- Estimated annual ROI
- Payback period
- Total annual revenue (generation × offer price)
- Basic feasibility score based on minimal parcel/community inputs

Rationale
- Interviews consistently show that communities cannot interpret energy economics.
- This feature establishes the baseline for all later decisions.
- Technically easy to implement (static modeling + simple visualization).

Success Criteria
- 80% of users can explain their community’s ROI within 1 minute.
- ≥70% users report increased confidence after using the dashboard.
- Sponsors can summarize the financial picture in under 30 seconds.

2. AI Chatbot (Explain + Guidance)

An AI chatbot that acts as a “clean energy guide” and supports:
- Plain-language explanations of terms (PPA, REC, IRR, incentives)
- Answers to community questions (e.g., “What does a 15-year term mean for us?”)
- Contextual coaching based on the dashboard (“Your ROI is low because…”)
- Quick “Should I worry about this?” risk clarifications

Rationale
- Replaces static Explain Mode with a conversational, human-like guide, which reduces confusion and boosts trust.
- Most requested usability enhancer from early interviews.
- First step toward a future “AI negotiation assistant.”

Success Criteria
- ≥60% users ask at least 3 questions during a session
- Users report chatbot increased understanding compared to reading alone

3. Contract Summary Generator

Upload a simplified PPA proposal and get an AI generates a plain-language summary:

- Key terms
- Parties' obligations
- Expected financial impacts
- Risks and benefits
- Simple red flags

Rationale

- Every interview identified contracts are too hard to understand.
- Low-cost, high-impact feature: template + AI summarization.
- Increases community confidence before negotiation.

Success Criteria

- 90% of users can paraphrase the core terms after reading the summary.
- Reading time is reduced by ≥60% compared to the original contract.
- Trust score ("I understand what I am agreeing to") ≥4/5.

4. ~~**Corporate Offer Comparison (Option A/B/C Cards)**~~

~~Visual comparison cards for multiple corporate solar offer options:~~

- ~~Each option includes ROI, risk notes, term length, REC inclusion~~
- ~~Simple recommendation label ("Best for short-term cash flow," etc.)~~

~~Rationale~~

- ~~Corporate buyers often present multiple PPA options.~~
- ~~Communities currently cannot evaluate which option is better.~~
- ~~Core value of SolarEase: early decision-support.~~

~~Success Criteria~~

- ~~Users consistently select the objectively best-fit offer (A/B test).~~
- ~~Confusion about offer options drops by ≥50%.~~
- ~~≥70% of test users use comparison cards to support internal discussion.~~

5. **Light Community Comparison Map (Deferred)**

A very lightweight map comparing:

- Each community's solar potential vs. 1 neighboring community
- Basic PPA price range (static)

Rationale

- Full multi-community modeling is too heavy for MVP.
- Sponsor and BoSong expect some gesture toward collaboration.
- Provides geographic context without complex engineering.

Success Criteria

- Users can describe collaboration potential after viewing the map (≥60%).
- At least 3 interview participants mention improved understanding of their region.
- Map loads in <5 seconds.

6. **Household Negotiation Tool (Now Included)**

A lightweight tool supporting internal community coordination when one household has higher generation, and one has lower solar potential, and communities need to agree on cost-sharing or benefit-sharing. MVP version includes:

- A simple interactive “preferences slider” (e.g., cost split 60/40, 50/50)
- AI chatbot mediating household trade-offs in plain language
- Output: a recommended “fair allocation summary”

Rationale

- This addresses household-level fairness, which BoSong emphasized.
- Keeps it simple (no complex game theory) but still meaningful.
- Shows the sponsor that you considered intra-community negotiation, not just corporate-community negotiation.

Success Criteria

- ≥60% users say the tool “helped clarify differences” between households
- Users can generate an internal agreement summary in <3 minutes

Optimization Tool/ Offer Comparison Tool

Community Profiles

Parameter	Unit	Example	Notes
Solar System Capacity	kW	500	Total installed capacity
Capacity Factor	decimal	0.18	Annual generation as % of capacity (region-specific)
Solar Degradation Rate	degradation_rate	0.005	Equipment efficiency decline (typically 0.5-1% annually)
Current Utility Rate	$/kWh	0.12	Community's current electricity cost
Utility Rate Escalation	decimal/year	0.03	Expected annual utility price increase
Discount Rate	decimal	0.06	Community's cost of capital for NPV calculation

PPA Option Parameters

Parameter	Unit	Example	Notes
Contract Term	years	10/15/20	Length of PPA

			agreement
PPA Rate	$/kWh	0.09	Initial PPA electricity price
PPA Escalation	decimal/year	0.02	Annual PPA price increase
Upfront Cost	$	5,000	Initial community payment (if any)
Annual O&M Cost	$/year	2,000	Operations & maintenance fee

Incentives

Parameter	Unit	Example	Notes
Federal Tax Credit Rate	decimal	0.30	ITC (Investment Tax Credit) - currently 30%
State/Local Grants	$	15,000	One-time grants
Tax Credit Eligibility	boolean	true/false	Can community claim tax credits?

Calculation

1. **Annual Solar Generation** (Calculate electricity generated each year, accounting for equipment degradation.)

Formula: generation[year] = capacity_kw × capacity_factor × 8760 × (1 - degradation_rate)^year
Where:
- 8760 = hours per year
- year = 1, 2, 3, ..., 20

2. **Utility Cost Avoided** (Calculate how much the community would have paid for grid electricity (the baseline for savings).

Formula:
utility_cost_avoided[year] = generation[year] × utility_rate_base × (1 + utility_escalation)^year

3. **PPA Cost** (Calculate actual cost to purchase solar electricity under the PPA.)

Formula:
ppa_cost[year] = generation[year] × ppa_rate_base × (1 + ppa_escalation)^year

4. **Net Annual Benefit** (Calculate actual net savings per year.)

Formula:
net_annual_benefit[year] = utility_cost_avoided[year] - ppa_cost[year] - om_cost_annual

For year 0 (upfront):
net_annual_benefit[0] = -upfront_cost + total_incentives

5. **Total Incentives Year 0** (Calculate one-time financial incentives received upfront.)

Formula:
tax_credit = upfront_cost × tax_credit_rate (if is_tax_eligible == true, else 0)
total_incentives = tax_credit + grants_total

6. **Cumulative Benefit** (Track when the community breaks even and total accumulated)

Formula:
cumulative_benefit[0] = net_annual_benefit[0]
cumulative_benefit[year] = cumulative_benefit[year-1] + net_annual_benefit[year]

7. **Net Present Value** (Discount future cash flows to present value for accurate comparison)

Formula (for each year):
npv_annual[year] = net_annual_benefit[year] / (1 + discount_rate)^year

Total NPV:
total_npv = Σ(npv_annual[0] to npv_annual[20])

Key Performance Metrics

1. **Total NPV Savings** (Higher NPV = Better financial return)

total_npv_savings = Σ(npv_annual[year] for year in 0 to contract_term)

2. **Return on Investment** (ROI)

total_undiscounted_benefit = Σ(net_annual_benefit[year] for year in 1 to contract_term)
roi = (total_undiscounted_benefit - upfront_cost) / upfront_cost

3. **Simple Payback Period** (How quickly the community recovers initial investment from first-year savings)

payback_period = upfront_cost / net_annual_benefit[1]

4. **First Year Net Benefit** (Shows immediate cash flow impact (important for communities with tight budgets)

first_year_benefit = net_annual_benefit[1]

5. **Total Long-term Value** (20-year)

total_20year_benefit = cumulative_benefit[20]

Information Hierchary

Homepage - V1

Possible Layout References: https://miro.com/app/board/uXjVJ4eIb3c=/

[Navigation Bar]

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Hero Banner

Hero Banner: Establish mission and emotional appeal

CTA: Start Simulation / Explore My Community

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My Community Snapshot (personal stats)

- Acts as the “personalized dashboard header” after login
- Key metrics: energy generated / Total savings / Carbon footprint / Applicable Policy*

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Dashboard with Map (Financial & Project overview)

- Dynamic map as the central visual element
- User can toggle between: Geographic view: EMC territories and nearby towns, Relational View: Partner networks
- Key overlays: Solar potential, ROI, Project phase icons, Partner Status, and REC eligibility
- AI agent sidebar

Collaboration & Group Simulation Section

- Visualize joint project potential when combining multiple communities
- Interactive comparison table: ROI uplift, fair-share distribution, shared PPA opportunities
- “Invite Partner Community” button, share simulation

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Policy & Incentive Spotlight

- Keep users informed about available grants, tax credits, or upcoming deadlines

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Negotiation & PPA Support Panel

- Centralized space for communities to view and simulate corporate PPA options
- Tabs include: public PPA summary, member negotiation rooms (requires login), leader review (restricted)

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Capital Project & Impact Dashboard

- Post-simulation, users track project development and impact metrics:
- Project phase: Planning → Construction → Commissioning → REC delivery
- Data points: Total installed MW/ Annual Generation/ CO2 offset/ Local job created
- Visualization: timeline bar, cumulative impact cards, plain-language summaries

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Save Simulations

- Allows users to revisit and compare prior results
- Each report includes:
- Inputs, assumptions, ROI, AI summary ("key learnings"), and export options (PDF/CSV/share)

PPA Support Page [To be updated]

[Home Page]

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[Bargaining Dashboard]

- **Public Summary of completed PPA data (read only)**

CTA "Join your community to participate in negotiation → [Sign in / Verify Role]"

- **Member Interactive Page (requires login)**

PPA Status update, adjust PPA indexes, external offer options

- **Leader Management (requires login/ restricted)**

Final PPA Draft

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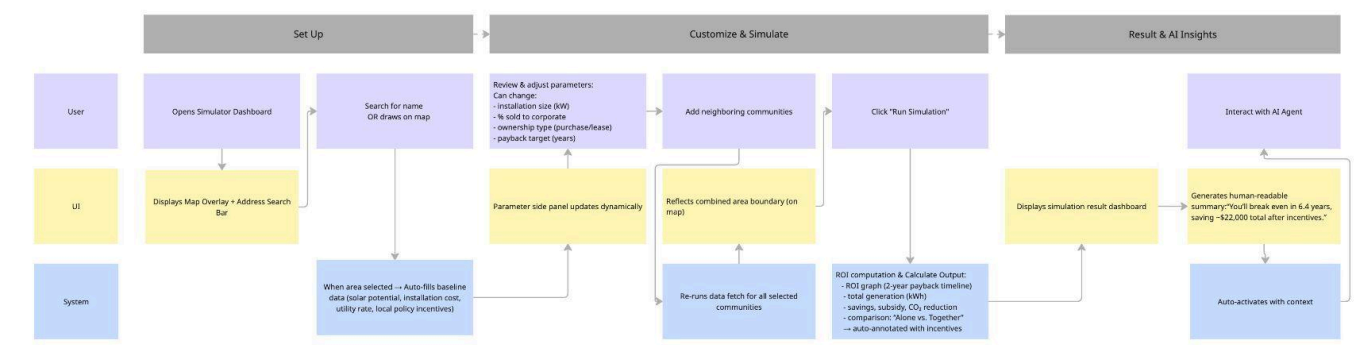
[Contract Library]

User Flow [To be updated]



User Journey [To be updated]

0) Landing

User action: Lands on dashboard, sees short value prop and "Select your community."
System: No calls yet. Shows an empty state with one sample case card.

1) Pick a location (point or polygon)

User action: Click a point on the map or draw a boundary → "Start Assessment."
System auto-fills (editable later):

- Geo: lat/lon, county, state, local utility/ISO zone.

- System defaults: initial capacity (e.g., 100 kW) or capacity from roof/parcel area (if user drew a polygon), default tilt/azimuth.

UI: Right panel shows “Preparing default assumptions…”
Backend: Reverse geocode + utility/ISO lookup. Kick off /autofill in background.

2) Auto-fill assumptions (from open data)
System auto-fills (all fields show source + timestamp, and are editable):

- **Generation: annual & monthly kWh, capacity factor, degradation**.
    - NREL PVWatts API: https://developer.nrel.gov/docs/solar/pvwatts/
    - Input: { lat, lon, system_capacity:100, tilt:20, azimuth:180 }
    - Output: annual & monthly kWh, capacity factor

- **Retail price/credit: local retail rate now, escalation assumption; or a target tariff/credit rule to derive a default Credit Rate**
    - EIA API: https://www.eia.gov/opendata/
    - Filter by state and sector=residential/commercial
    - Get latest price (¢/kWh)

- **Incentives: federal Investment Tax Credit, applicable state/local incentives, SREC applicability & a default price**
    - Use DSIRE API (need to pay); or:
        - Mirror DSIRE CSV/SPARQL export in Google Sheet or Postgres table (ETL weekly).
        - Query by state and technology = “solar photovoltaic”.
        - Return: ITC:xx%, state_incentives, SREC_applicable.

- **Costs & O&M: capex $/W baseline, interconnection/soft costs, O&M $/kW-yr**
    - Download NREL ATB dataset (CSV) once; store locally or in Postgres.
    - Query by region = state or NERC region → get capex_per_W, opex_per_kW_year.

- **Financials: Discount rates, loan ratio, loan interest, analysis years**
- **(Optional) Commercial terms: default PPA price range (or bill-credit proxy), subscription structure defaults (anchor share, ramp years)**

UI: A collapsible “Assumptions” drawer is populated, each row shows a “lock/unlock / restore default” control. A “Calculate” button appears (or auto-calculates).
Backend: POST /autofill {lat, lon, capacity_kW, tilt?, azimuth?} → unified JSON.

3) One-click simulate (single community)
User action: Press “Simulate.”
Inputs used: Location-derived generation; retail/credit assumptions; incentives; costs & O&M; finance; commercial/PPA settings; any user edits.
UI outputs (Results panel):

- Top cards: Payback (yrs), IRR, NPV, 1st-year net cashflow.
- Scale & energy: capacity, annual kWh, utilization hours.
- Revenue breakdown: PPA/credit, incentives, SRECs, other.
- Cost breakdown: CAPEX, OPEX, interconnection, soft.
- Charts: cumulative cashflow, stacked revenue, small sensitivities (+/-10%).
- AI Explain sidebar: plain-English rationale and next-step hints.

Backend: POST /simulate → ROI engine returns KPIs and breakdowns.

4) Tweak the big levers (iterate)
User action: Adjust 2–3 high-impact sliders/inputs (PPA or Credit Rate, capacity, discount rate, capex $/W, anchor share).
UI: Realtime recompute; a small delta banner shows “vs. initial plan: IRR +1.3pp / Payback –0.8 yrs.”
Backend: Repeat POST /simulate with changed fields (cache & partial recompute to keep <300 ms if possible).

5) Multi-community aggregation (optional, MVP+)
User action: Select neighboring communities on the map → “Add to Group Simulation.”
System: Bulk autofill for each selected community; share global assumptions (PPA, discount), keep local differences (capacity, roof area, interconnection).
UI outputs:
- Totals: summed capacity/kWh, group cashflow, group IRR/NPV.
- Uplift: “Adding Community B → group IRR +0.7pp (if PPA ≥ $X/MWh).”
- Sharing: compare fair-share rules (by energy, by capex, hybrid).

Backend: POST /autofill:batch, then POST /simulate:aggregate.

6) Transaction scenarios (optional, MVP+)
User action: Open “Corporate Off-take” tab; move a PPA price / split slider; or switch to “bill-credit program” mode.
UI: Two-sided summary (community annual benefit vs. buyer annual cost / clean MWh locked), contract bullets, risk flags.
Backend: Same POST /simulate, with scenario:"ppa" or "bill-credit".

7) Save / export / share
User action: Save scenario, export PDF/CSV, copy share link.
UI: Exports include inputs (with sources), KPIs, charts, and a human-readable summary.
Backend: POST /scenarios, GET /export?format=pdf|csv.

Technical Details

Financial Model input & output

Input

Generation	Installed system capacity, Monthly energy generation, Annual energy

	generation, Annual degradation rate
Retail price/credit	PPA selling price, Bill credit rate, Retail electricity price, Annual price escalation rate
Incentives	Federal Investment Tax Credit rate, State or local incentive amount, SREC price, Production-based incentive, One-time rebate
Cost	Capital cost per watt, Soft costs (permits, admin, etc), Interconnection cost, Annual O&M cost
Financials	discount_rate, loan_ratio, loan_interest, analysis_years

Output

Financials	Net Present Value, Internal Rate of Return, Payback period, First-year net cashflow
Energy performance	Annual generation, Capacity factor
Revenue & cost composition	Revenue breakdown, Cost breakdown
Timeline view	Cashflow over time

Comparison with Cohort 7 work

Dimension	**Single-community**	**Multi-community**
User	One community	Multiple communities simultaneously
Focus	Assess ROI for a single site	Enable collaboration and aggregated selling
Economic model	Independent ROI calculation	Aggregation model (joint PPA, shared profit)
Dashboard display	Focused on one community's data	Comparative and collective view across communities
Goal	"Should we invest in solar?"	"How can we earn more together and strengthen our voice?"

Reference

[1] Quincy Population: https://censusreporter.org/profiles/40000US73072-quincy-wa-urban-area/
[2] East Wenatchee Population: https://datausa.io/profile/geo/east-wenatchee-wa
[3] Malaga Population:
https://www.point2homes.com/US/Neighborhood/WA/Malaga-Demographics.html